
PRIVYSE

On-device Privacy-Preserving Browser Agent

SIH 26171 · Product Pitch & Demo Script

One-line pitch. “PRIVYSE is a browser agent that **browses the web for you** — while proving, on your own device, that **your personal data never leaves it.**”

1 The Problem

- **Generic AI “computer use” leaks everything.** Off-the-shelf browser agents capture the raw screen — every email, phone number, PAN, address, shortlisted result — and ship those pixels to a hosted model. The user “wins” convenience and loses privacy.
- **The screen is a firehose of personal data.** A single viewport can contain credit-card numbers, Aadhaar/PAN IDs, face photos, OTPs and addresses passim — all in one screenshot.
- **Government and enterprise touchpoints are sensitive.** Forms for admissions, applications, banking and PDS are exactly the pages where an agent is most useful *and* most dangerous.
- **Trust must be demonstrated, not promised.** A privacy claim without visible proof is marketing; judges and users need a *live, on-device gate* they can watch operate.

2 The Solution: A Privacy Gate the User Can Watch

PRIVYSE puts a **sanitizer** between the camera and the model. Every screenshot passes through a five-stage pipeline before anything can leave the device:

CAPTURE	SANITIZE	GATE	REASON	ACT
screen grab	detect + redact	zero-leak OCR	local VLM	click / type

The gate is fail-closed. If the verification layer cannot run, the step is blocked — a message is never sent “best-effort unsanitized.”

Redaction is tiered on-device:

- **Tier A — absolute secrets.** Password, OTP, CVV, API-key fields: solid black boxes, no text, never readable.
- **Tier B — personally identifiable information (PII).** Emails, phones, Aadhaar, PAN, Luhn-valid cards, names, addresses: black-boxed *and* replaced with stable tokens ([EMAIL_1], [PAN_2], ...). Same value $\Rightarrow$ same token, so the model keeps continuity across steps.
- **Tier C — faces & non-DOM text.** MediaPipe BlazeFace runs on-device to blur faces via stacked downscale/upscale; Tesseract OCR scans only image/canvas/video regions the DOM never carries.
- **Sensitive pages are never captured at all.** Banking, password-reset, and government-ID pages skip the screenshot entirely.

- **Zero-leak OCR verification.** The *sanitized* image is OCR'd again: if any PII is still readable in the pixels that would be uploaded, the gate reports BLOCKED and the step stops.

3 Product Surface

3.1 Side panel — the control center

A cockpit opened with one click: task box, live agent-state pill, the 5-stage pipeline strip with a GATE node, a privacy hero card (detected / protected / raw-values-sent, with the stable-token map shown as `jes***@gmail.com` → `[EMAIL_1]`), a *before/after* “Your screen vs. What the AI sees,” a structured decision trace, a live activity log with human-readable lines, and a foldered session history with per-run privacy summaries.

3.2 Spotlight (Alt+K)

A page-level command palette: run a task, step once, stop, toggle the cursor, switch models, check VLM health, or clear learned lessons — without leaving the tab.

3.3 Floating PRIVYSE orb

When the panel is closed, a draggable status dot floats on the page; during a run it expands into a live pill (stage + step counter + pipeline dots) with a green GATE PASS / red BLOCKED flash per step. Clicking it re-opens the panel. It hides itself during every capture so it never appears in a screenshot the model sees.

3.4 Privacy Lens

A one-key scan highlights sensitive regions *live on the page* — red rings with type tags (`email`, `pan`, `card`, `address`) — plus a banner offering **Show sanitized view**.

3.5 AI View mode (USER | AI VIEW)

Flips the page into exactly what the model sees: PII rewritten to stable tokens and Set-of-Marks chips on every interactive element, numbered by the same walk as the SoM overlay the VLM is given. Judges literally read the agent’s viewport.

4 Trust & Safety Guarantees

- **Everything sensitive happens on-device.** Detection, tokenizing, face-blurring, OCR verification — all in the extension. The server receives only the sanitized JPEG and the tokenized DOM snapshot.
- **Raw values are never uploaded.** The outbound payload is scanned (`assertNoLeaks`) before sending; zero raw values sent is a live (and visible) metric.
- **Fail-closed everywhere.** Vision host down → degrades to DOM-only redaction; the layer-2 OCR gate unavailable → the DOM-level gate still enforces; a blocked step is never emailed out.
- **Visible receipts.** Every step logs what was redacted, which tokens mapped from which masked values, and the gate verdict.

5 Tech Stack

Layer	Choice	Why the judges care
Extension	WXT · Chrome MV3	modern, fast, single manifest
On-device vision	MediaPipe BlazeFace + Tesseract.js (WASM)	faces blurred / image-region text redacted without a server
Orchestration	MV3 service worker + offscreen vision host	rate-limit-aware capture, warm model preload
Agent loop	adaptive: /act + /re-think, stuck / no-progress detection	never spin forever; self-recovers
Local VLM	Ollama Qwen2.5-VL via FastAPI (SSE stream)	model choice is dropdown-selectable
Recovery layer	site-level lesson memory, auto-type escalator	the demo rarely gets “stuck” in front of a judge

6 Live Demo Script (60–90 seconds)

1. **Open a real form page** (application / checkout / search with a visible email + phone already typed in). *“Every screenshot of this page contains personal data. Watch what happens to it.”*
2. **Press Alt+K, type a task, run.** Spotlight launches the loop right on the page.
3. **Watch the pipeline light up.** CAPTURE → SANITIZE → GATE → REASON → ACT on the panel — call out each stage as it fires.
4. **Freeze on the privacy hero.** “ N detected, N protected, 0 raw values sent.” Point at a token row: `jes***@gmail.com` → `[EMAIL_1]` (the masked form is the *only* representation of that value that has ever been shown).
5. **Open the before/after card.** “Your screen — what the AI sees.” The model’s viewport is clean.
6. **Enable Privacy Lens** on the live page. Red rings + type tags appear over the exact regions just protected.
7. **Toggle AI VIEW.** The page itself becomes the model’s view: tokens in prose, SoM chips on every interactive element. *“This is literally the pixel stream the model reasons over.”*
8. **Close the panel — the orb takes over.** It tracks the run live; click it to reopen the cockpit.
9. **End on a gate pass.** Green GATE PASS; task complete, decision trace shows the goal / observed / protected / target SoM / action / result.

Close-out line. “Work is done when the task is done — and the person whose data was on that screen *saw* that nothing left the device. That is the difference between “*trust us*” and “*watch us*.” ”

7 Why PRIVYSE Stands Out

- **Provable privacy, not perceived.** A live, judge-visible gate with receipts on every step.
- **Fully on-device redaction.** No private pixels ever travel; the VLM is local and optional/swapable.
- **Works on *real pixels*, not just DOM text.** Faces, canvases, videos and image regions are covered — the failure mode generic agents ignore.

- **Product, not a script.** Side panel, spotlight, orb, lens, AI View, histories, model selector — a complete user experience out of the box.
- **Self-growing.** The agent learns site-level lessons from its own mistakes and never repeats them.

PRIVYSE · SIH 26171 · “Your screen never leaves — and you can watch it.”